# Javad Komijani

*Research Scientist | Generative Models & Statistical Inference*

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## Professional Summary

Theoretical and computational physicist working on physics simulations, statistical inference, and generative modelling. I develop research ideas end to end, from mathematical formulation and algorithm design to PyTorch implementation and computational experiments. My distinctive expertise, rooted in effective field theories (EFTs), is identifying the collective variables that drive large-scale behaviour in complex systems, and using them to interpret data and to construct efficient quantitative models with controlled uncertainties. Notably, I independently developed the theory, EFT model, and statistical analysis for precision quark masses.

In machine learning, I combine this perspective with equivariance to develop efficient generative models, and I am interested in how these principles can inform interpretability, distillation (matching an effective model to a more complete description), and the understanding of scaling behaviour.

**Research impact:** 27 peer-reviewed publications, h-index 20, 3,500+ citations (Google Scholar, Nov 2026).

## Selected Research Contributions

- **Precision lattice QCD and Standard Model parameters**: Combined three effective field theories into a single effective description, used as a statistical model that extrapolates expensive simulations to physical limits with a full error budget. Key contributor to high-precision determinations of quark masses and meson decay constants, with results incorporated into the world reference averages of the Particle Data Group. Introduced a new quark-mass scheme (MRS) that removes the leading divergence of perturbation theory without an arbitrary cutoff, improving the precision of quark-mass determinations. 10+ papers; e.g., Phys. Rev. D 98, 054517 (2018) (quark masses); Phys. Rev. D 98, 074512 (2018) (decay constants).
- **Diffusion models on Lie groups**: Developed diffusion models for SU($N$) gauge theories with a Hamiltonian-dynamics corrector; co-developed group-equivariant diffusion models. arXiv:2605.06134 (JHEP, accepted); JHEP 07, 213 (2026).
- **Sampling with exact correction** (single author): Factorized 2D SU($N$) gauge theory into independent plaquettes and a single global variable. Sampled the plaquettes with normalizing flows and the global variable with a conditional normalizing flow, followed by an exact reweighting correction. Sampling is unbiased, fully parallel, and free of Markov-chain burn-in; the flows act on local objects, so training is cheap and does not grow with lattice size. Phys. Rev. D (accepted), arXiv:2610.09147 | notebooks

## Research Experience

2021–2026 **Postdoctoral Researcher**, *ETH Zurich*

- Research on generative models for lattice field theory: diffusion models, normalizing flows, flow matching, and flow maps on Lie groups, from derivation to PyTorch implementation, distributed training, and statistical validation.
- Designed parameter-efficient architectures: an EFT-inspired power-spectral-density flow, and an SVD-based gauge-equivariant flow whose 1,288-parameter block (PoS Lattice 2024) matched the effective sample size of a much larger published spectral flow.
- Integrated normalizing-flow modules into the EU interTwin Digital Twin Engine.
- Co-led lattice-QCD simulation and analysis projects with PhD and MSc students; co-PI on successful compute-time proposals at the Swiss National Supercomputing Centre (CSCS).

2015–2021 **Postdoctoral Researcher**, *TU Munich / Glasgow / Tehran*

- Precision lattice-QCD analyses with EFT-based Bayesian fits and full error budgets.
- Large-scale simulation and analysis pipelines on HPC systems in major lattice-QCD collaborations, such as MILC and Fermilab Lattice.

## Open-Source Software

lattice_ml **PyTorch library for generative modelling and sampling on Lie groups**

- Diffusion models with predictor–corrector samplers, flow matching, and flow maps on Lie groups; hybrid Monte Carlo for U(1) and SU($N$).
- Adjoint-method ODE solvers for memory-efficient backpropagation; symplectic integrators; equivariant layers; distributed training.

normflow **PyTorch package for normalizing flows**

- Training by reverse KL (with path gradients) or from data; Metropolis-corrected sampling.
- EFT-inspired power-spectral-density flows; spectral and SVD-based gauge-equivariant flows; coupling, spline, and autoregressive layers.

## Technical Skills

ML Diffusion and score-based models, flow matching, normalizing flows, equivariant architectures, U-Net and transformer backbones, automatic differentiation

Theory & Statistics Quantum and statistical field theory, effective field theory, Bayesian inference and uncertainty quantification, Markov chain Monte Carlo, importance sampling, asymptotic analysis, Lie groups

Computing Python, PyTorch on GPUs, JAX, distributed training; C++; HPC systems; Linux, Git

Languages English (fluent), German (basic, A2)

## Mentoring & Leadership

Mentoring **Mentored and co-supervised researchers in physics & ML:**

- PhD students mentored (3): gauge-equivariant diffusion models; Fourier-accelerated sampling for SU($N$); statistical analysis of lattice-QCD data.
- MSc theses co-supervised (6): equivariant U-Net; autoregressive, continuous-time, and spline-based flows; simulated tempering.
- BSc and intern projects: path-integral Monte Carlo; CNNs in arbitrary dimensions.

Organization Co-organized five workshops, including ZPW 2026 (ML Techniques for Particle Physics) and ML Lattice 2025.

Refereeing Referee for Physical Review D and other physics journals (10+ reviews since 2016), including machine-learning papers.

## Education

2015 **PhD in Physics**, *Washington University in St. Louis*
Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics* (advisors: C. Bernard, C. M. Bender). EFT-based frameworks and Bayesian inference for lattice QCD; nonlinear eigenvalue problems.

2006, 2009 **BSc and MSc in Electrical Engineering**, *University of Tehran*
MSc focus: electromagnetic waves and fields, signal processing, statistical analysis.

## Selected Publications


2026 J. Komijani, "Sampling SU($N$) gauge theory on a 2D lattice from independent plaquettes via holonomies and corner reweighting," Phys. Rev. D (accepted), arXiv:2610.09147.

2026 J. Komijani, M. K. Marinkovic, L. Turgut, "Diffusion model for SU($N$) gauge theories," JHEP (accepted), arXiv:2605.06134.

2026 O. Vega, J. Komijani, A. El-Khadra, M. K. Marinkovic, "Group-equivariant diffusion models for lattice field theory," JHEP 07, 213 (2026).

2025 J. Komijani, M. K. Marinkovic, "Normalizing flows for SU($N$) gauge theories employing singular value decomposition," PoS Lattice 2024, 050, arXiv:2501.18288.

2023 J. Komijani, M. K. Marinkovic, "Generative models for scalar field theories: how to deal with poor scaling?" PoS Lattice 2022, 019, arXiv:2301.01504.

2018 A. Bazavov *et al.*, "Up-, down-, strange-, charm-, and bottom-quark masses from four-flavor lattice QCD," Phys. Rev. D 98, 054517 (2018).

2018 A. Bazavov *et al.*, "$B$- and $D$-meson leptonic decay constants from four-flavor lattice QCD," Phys. Rev. D 98, 074512 (2018).